

MECH 151

# Technical Design

## Unit Reference Guide

*A student quick-reference for every unit — Design Fundamentals through Design for Manufacturing — for portfolio work, quizzes, and the final prototype.*

Foundations

Unit 1

Unit 2

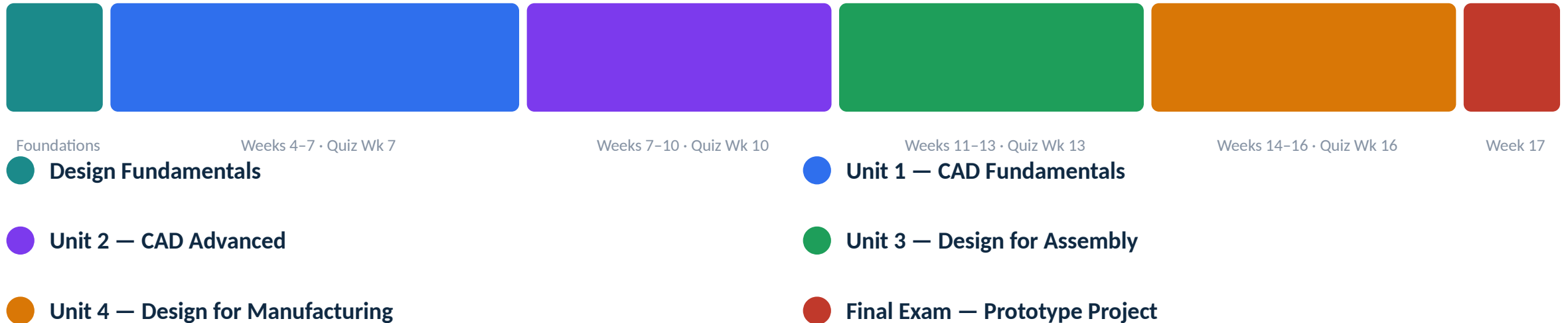
Unit 3

Unit 4

Final Exam

# Semester Roadmap, Weeks 4–17

*One project-based portfolio module per week; a short foundations unit precedes Unit 1.*



**How grading lines up:** each unit maps to a graded portfolio module and closes with a unit quiz; the Design Portfolio (50% of the course grade) and the Prototype + Final Exam (30%) are graded separately — see the syllabus for the remaining weighting.

# This Course Is CAD-Program-Agnostic

*The field of CAD software is too broad to teach every package — so this course teaches the operations that transfer between all of them.*

**Dozens of parametric CAD packages** — are used across industry — SolidWorks, Fusion 360, Onshape, Inventor, Creo, NX, CATIA, and more

**It's impractical to teach all of them in one course** — and mastering only one would leave you stuck the moment a job, internship, or team uses something else

**So this course teaches the operations every one of them shares** — the specific menu names are yours to look up in whatever software you're using

| The Operation                         | Sometimes Called                             |
|---------------------------------------|----------------------------------------------|
| Push/pull a 2D profile into a solid   | <i>Extrude • Pad • Boss / Cut</i>            |
| Spin a profile around an axis         | <i>Revolve • Rotate</i>                      |
| Lock a sketch so it can't drift       | <i>Relations • Constraints</i>               |
| The running history of modeling steps | <i>Feature Tree • Timeline • Model Tree</i>  |
| Position parts relative to each other | <i>Mates • Joints • Assembly Constraints</i> |
| Round over or angle-cut a sharp edge  | <i>Fillet / Chamfer • Round / Bevel</i>      |

**Why it matters:** everything in this deck — sketching, feature trees, mates, GD&T — describes the operation, not a specific menu path. Whatever software you're using in class, an internship, or a job, look for the tool that does the same job.

FOUNDATIONS

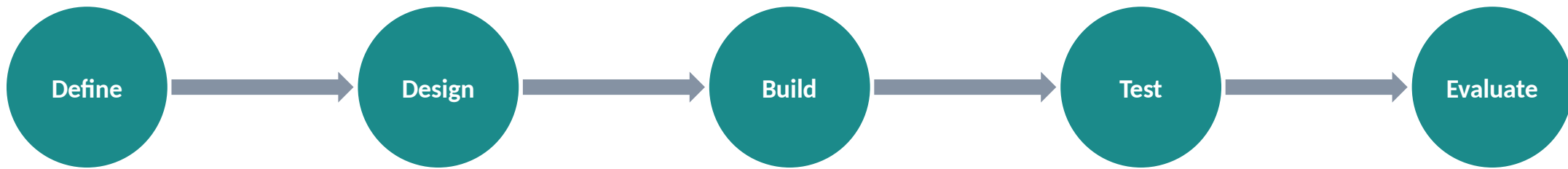
# Design Fundamentals

*Step back from the software: what a CAD model is for, before Unit 1's sketches and features begin.*

BEFORE UNIT 1

# The Design Process Is a Cycle

*Every design loops — the first prototype is expected to be wrong, and that's the process working.*



*↺ then back to Design (or Define), carrying what you learned*

**Define** — state what the part must do, fit, withstand, and cost — before any geometry exists

**Design** — concept options, then the parametric models and drawings that pin down every dimension

**Build** — make a prototype — a print, a machined part, a mock-up

**Test** — measure it, fit it, load it, use it the way it will really be used

**Evaluate** — compare what happened against the requirements; loop back with what you learned

**Why it matters:** if you assume the first version is final, you build a model that's painful to change. If you assume you'll revise it, you build for iteration — clean sketches, captured intent, a readable feature tree.

# Define the Problem Before You Design

*The step skipped most often — and the one that wastes the most time when skipped.*



## Specific

says exactly what, for whom



## Measurable

has a number you can check



## Achievable

possible with your time & tools



## Relevant

actually serves the need



## Time-bound

due by a defined point

**SMART in practice:** "hold a phone at a comfortable viewing angle" is a direction. "Hold a 600 g phone at 55–70° without tipping, printable in under 4 hours, done by Week 12" is something you can design to and test against.

- Requirements, objectives, and constraints get lumped together as "the spec" — but they filter and rank designs differently (next: how each one works).

# Requirements, Objectives, Constraints

*Requirements and constraints filter designs to the acceptable ones; objectives rank the acceptable ones to find the best.*

## Requirements

*What the design must do — pass/fail*

- "Supports 5 kg without permanent deflection"
- Either met or not; missing one disqualifies a design
- Verified by test or inspection

## Objectives

*What you're optimizing — directions of "better"*

- "As light as possible," "minimize cost"
- Not pass/fail — compared by degree
- Often trade against each other

## Constraints

*Fixed limits on the solution space*

- "Fits a 150 mm cube," "\$25 budget," "due Week 12"
- Not about function — about what's allowed
- Violating one disqualifies the design

**Why it matters:** mix them up and you either reject a good design for missing a "nice to have," or ship one that fails a real must-have.

**Evaluating success** — turn objectives into weighted, measurable criteria and score concepts with a decision matrix — screen pass/fail items first, then rank.

# Knowing When to Stop: The 80/20 Rule

*No design is ever perfect — chasing perfection is its own failure mode.*

## 80/20

~80% of a design's useful performance comes from the first 20% of the effort — chasing the rest costs far more than it returns.

**Move forward once requirements are met** — even if you can see ways it could be nicer

**Perpetual designing is a cost** — every revision delays the prototype, the test, and everything you'd learn from them

**"Good enough" is a decision** — judge against written requirements, not an imagined ideal

**Hold both ideas at once:** expect the first version to be wrong, and resist the urge to perfect it. The cycle moves forward when a design is good enough to learn from — not when it's flawless, and not before it meets its requirements.



# Design Intent

*The reasoning a model should capture — what you expect to stay true when a dimension changes.*

**Sketch relations** — symmetric, equal, concentric, collinear say why geometry is where it is, so it stays right when dimensions change

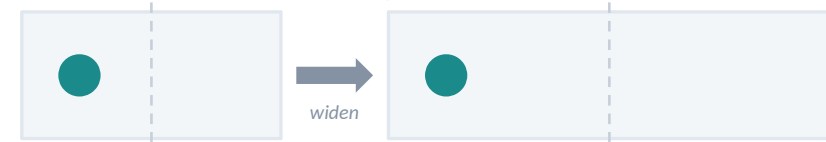
**Which dimensions you choose** — dimension from the reference that actually matters — a mounting face, a mating edge — not whatever's convenient

**Feature order & references** — build later features off stable geometry, not faces that may move or disappear in an edit

**Equations & linked values** — when two dimensions must always match or follow a rule, drive one from the other instead of typing both

*Same result, different intent — dimension a hole two ways:*

**Dimensioned from the left edge:**



→ widen the block and the hole drifts off-center

**Tied to center by a symmetric relation:**



→ widen the block and the hole stays centered

**Why it matters:** both models look identical until the block changes size — only the second one captures what you actually meant, not just a position that happened to look right.

1

**Editability**

a well-built model updates the way you intend

2

**Communication**

the next person can read the tree and see the logic

3

**Downstream reuse**

assemblies, drawings & manufacturing files inherit it

# Organizing a Design

*When a design has more than one part, where do the controlling dimensions live?*

## Bottom-Up

*Design each part alone, then assemble*

- Parts are independent — editing one doesn't ripple
- Simple to manage; easy to reuse parts
- Good for well-understood or off-the-shelf parts
- Weak when parts must share dimensions

## Top-Down

*Start from the assembly's shared geometry*

- Shared dimensions live in one place and propagate
- Keeps mating parts fitting as the design evolves
- Good for new designs with moving interfaces
- More setup; unmanaged references get fragile

**Tools for top-down** — layout sketch (fixes key sizes before parts exist), skeleton/master model (shared reference geometry), in-context references (shape one part against another directly).

**Rule of thumb:** use the simplest structure that captures the intent — reach for top-down only where parts genuinely must share dimensions.

# Practices That Keep a Design Robust

1

## Fully define every sketch

an under-defined sketch can drift on an unrelated edit

2

## Reference stable geometry

the origin and default planes never move; a filleted edge might vanish

3

## Keep sketches simple

one idea per feature — complexity goes in the feature tree

4

## Name & organize features

a tree of "Extrude7, Cut12" tells the next person nothing

5

## Minimize dependency chains

fewer things a feature relies on means fewer ways it breaks

6

## Model the design, not just the shape

if a rule governs the part, build the rule in

UNIT 1 • WEEKS 4-7

# CAD Fundamentals

*Sketching, views, and the first parametric model — the foundation every later unit depends on.*

QUIZ IN WEEK 7

# There Is No Single "Right" Model

*The same shape can be built from a different base feature, sketch set, or feature order — with effectively infinite valid combinations.*

**What actually separates good from poor:** two classmates can hand in completely different feature trees and both be correct. What separates a good model from a poor one is whether it captures design intent, rebuilds without errors, and stays easy to edit — not whether it matches a canonical answer.

**Focus on** — making deliberate choices you can explain, not on guessing the "expected" answer.

**This unit builds** — the drawing and modeling foundation every later unit depends on — a controlled sketch, standard views, and a parametric solid that updates when you change it.

# The Origin: Where Every Model Begins

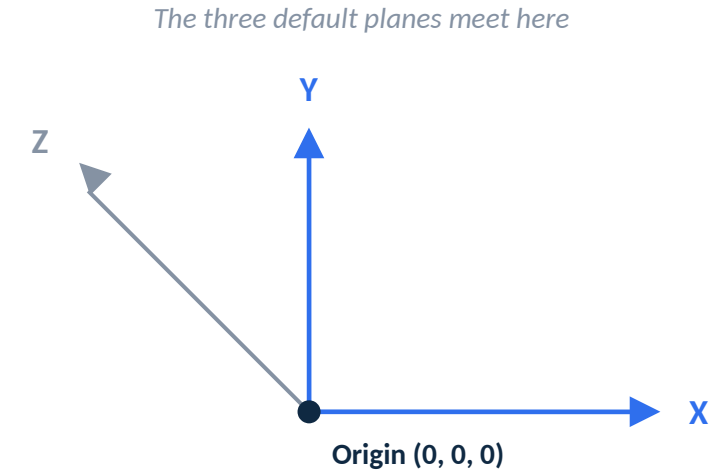
*Every coordinate, sketch, and feature ultimately measures from one fixed point in space.*

**The origin (0, 0, 0)** — the one point in the file that never moves, no matter how the model is edited

**Three default planes** — Front, Top, and Right all pass through the origin and split space along the X, Y, and Z axes

**Everything is measured from it** — directly or indirectly — a sketch plane, a mate, even a drawing's projected views all trace back to this fixed reference

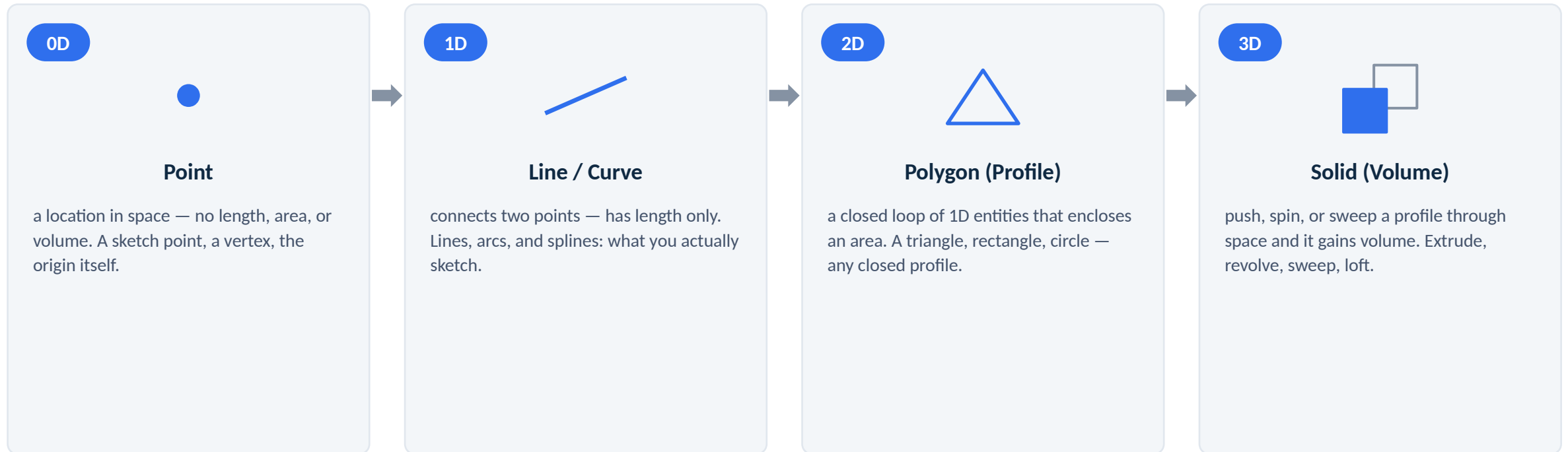
**Why it matters** — reference the origin (or geometry built from it) instead of an edge that might move, get filleted away, or get deleted in a later edit



**Why it matters:** the exact pairing of planes to axes (which plane is "Front") varies by CAD program — but every parametric system anchors its geometry to an origin and a set of default planes the same way.

# From Point to Solid: Building Up in Dimensions

*Every model is a stack of dimensions — each one built from the last.*



**Coming up:** the next several slides walk through sketching (0D → 1D → 2D) and solid modeling (2D → 3D) in detail — this is the map for where each one fits.

# Creating a Body: The Feature Loop

*Every solid feature repeats the same four steps — the feature tree is just this loop, run again and again.*



*↺ repeat, starting the next loop from a face this feature just created*

- 1 • Pick a plane** — one of the three default planes, a flat face of the solid so far, or an offset/angled reference plane
- 2 • Define a profile** — sketch a closed, fully-defined 2D profile on that plane — points and lines built into a polygon
- 3 • Create a volume** — extrude, revolve, sweep, or loft the profile into a solid, then add, cut, or intersect it with what's already there
- 4 • Dress it up** — apply fillets, chamfers, shells, drafts, or patterns — operations on the solid, not a new sketched volume

**Why it matters:** step 1 of the next loop is usually a face the previous loop just created, not a default plane — that dependency is exactly what builds the feature tree, one branch at a time.



# Sketching Fundamentals

*A sketch is a 2D profile on a plane or face — the input to a 3D feature like an extrude or revolve.*

1

## Sketch entities

lines, arcs, circles, rectangles, slots, splines, and points

2

## Reference geometry

the origin, the three default planes (front, top, right), and construction lines that guide without becoming part of the profile

3

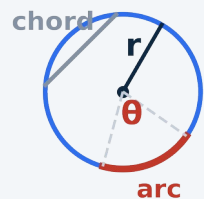
## Closed vs. open profiles

a closed loop can be extruded into a solid; an open profile makes thin features, surfaces, or paths

**Why it matters:** keep sketches simple — draw only the geometry the feature needs, keep profiles closed, and put complexity into separate features rather than one crowded sketch.

# Geometry of Shapes: 2D Construction

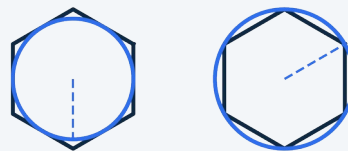
Before a shape is a sketch, it's plane geometry — points, lines, and curves in defined relationships.



## Circle & Arc Geometry

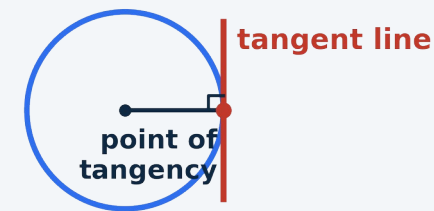
center + radius/diameter; a chord connects two points, a central angle subtends an arc.

Inscribed (flat-to-flat)      Circumscribed (point-to-point)



## Polygons

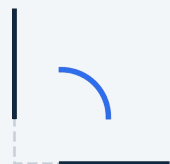
inscribed (flat-to-flat) vs. circumscribed (point-to-point) — matters for hex features and bolt heads.



## Tangency

a line/arc touches a circle at exactly one point, perpendicular to the radius there.

Fillet



Chamfer



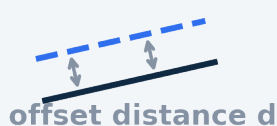
## Fillets & Chamfers

a fillet is a tangent arc replacing a sharp corner; a chamfer is a straight cut across it.

Bisector



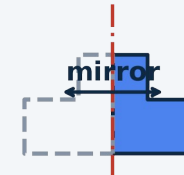
Offset (Parallel Copy)



## Constructions

perpendicular bisector and offset — classic constructions, now done by relations.

centerline



## Symmetry & Centerlines

build half a shape about a construction line and mirror it to guarantee identical halves.

# Sketch Constraints & Relations

*Constraints lock down where sketch geometry can move, removing degrees of freedom until the sketch is fully defined.*

## Under-defined

- Geometry can still be dragged
- The sketch is not fully controlled

## Fully defined

- Every entity fixed by relations + dimensions
- The target state before finishing a sketch

## Over-defined

- Conflicting or redundant constraints
- Must be resolved, not ignored

**Geometric relations** — coincident, collinear, parallel, perpendicular, tangent, concentric, equal, horizontal, vertical, symmetric

**Dimensions** — linear, radial, diameter, angular

**Why it matters:** add relations before dimensions — relations capture intent (this hole stays centered) so changing one dimension updates the model predictably instead of distorting it.

# Orthographic Projection & Multiview Theory

*Imagine the part inside a glass box — each face of the box catches one view; unfolding it gives the standard view arrangement.*

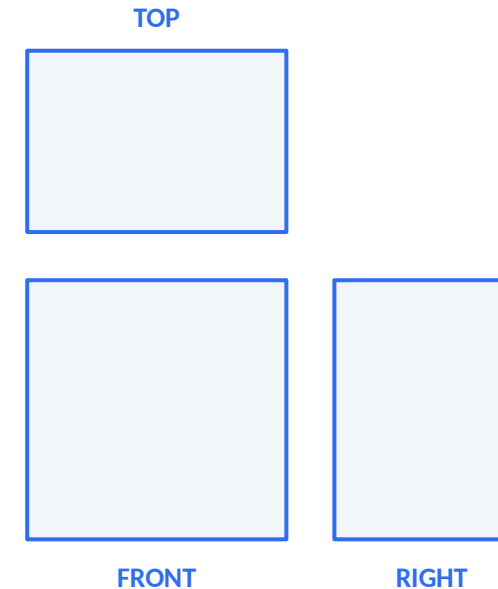
**Third-angle projection** — the U.S. standard — top view sits above front, right-side view sits to its right

**Line conventions** — visible edges solid, hidden edges dashed, centerlines for holes and symmetry axes

**View selection** — choose the front view with the most shape and fewest hidden lines; add only the views needed

**Descriptive geometry** — true length appears only when a line is parallel to the viewing plane; true shape only in a view perpendicular to that plane

**Primary vs. secondary views** — a primary view projects from a principal view; a secondary view projects from a primary view to reveal a feature still not true-size



# Basic Dimensioning Standards

*Dimensions tell the reader how big the part is and how precisely each feature must be made (ASME Y14.5 in the U.S.).*

1

## Anatomy

extension lines, dimension lines, arrowheads, and the value — kept off the outline and out of the view

2

## Placement

dimension to visible outlines, not hidden lines; group related dimensions; smallest closest to the view

3

## Don't over-dimension

each feature is located once — a redundant dimension creates conflict on the shop floor

4

## Feature callouts

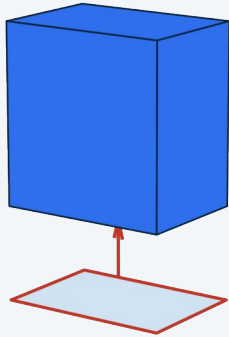
diameter, radius, counterbore, countersink, and depth symbols for holes

# Geometry of Shapes: Building 3D Solids

*Every solid model is a running total: start with a volume, add more, and cut some away.*

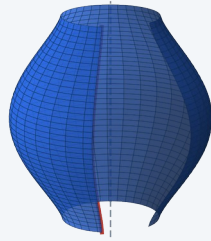
## Extrude

push a profile along an axis



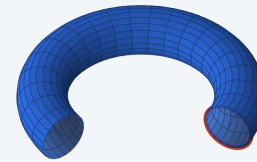
## Revolve

spin a profile around an axis



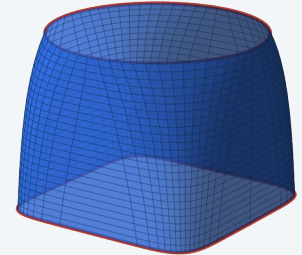
## Sweep

run a profile along a path

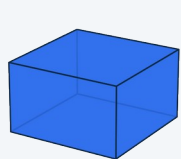


## Loft

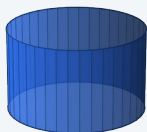
blend between different profiles



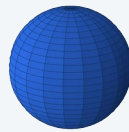
## Primitives



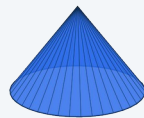
Block



Cylinder



Sphere



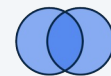
Cone



Torus

## Booleans & Dress-Up

### Union



combine A + B

### Subtract



remove B  
from A

### Intersect



keep only  
overlap

### Shell



hollow, keep  
wall t

### Draft



taper wall  
by angle

### Pattern



repeat at  
spacing d

*Fillet & chamfer are shown on the sketching slide (2D construction).*

# Introduction to Parametric Part Modeling

*A parametric model is a history of features, each driven by dimensions and relations — change a dimension and the model rebuilds.*

**Base feature** — the first solid, usually an extrude or revolve of a fully defined sketch

**Added features** — extrude, cut, revolve, fillet, chamfer, hole, shell, pattern — each added in order on the feature tree

**Sketch planes & references** — choosing where each feature's sketch lives so edits stay stable

**Practice** — editing driving dimensions, reordering/suppressing features, fixing rebuild errors — reading a feature tree and predicting what an edit will do

**Why it matters:** this is what makes CAD editable rather than something you redraw from scratch.

# Geometry of Shapes: Measurable Properties

*Every shape carries numbers that follow from its geometry — the values a design is checked against.*

## Of a 2D profile

- Area, perimeter, centroid
- Second moment of area ( $I$ ) — resistance to bending
- Section modulus ( $I \div \text{distance to extreme fiber}$ )

## Of a 3D solid

- Volume, surface area
- Mass (volume  $\times$  density), center of gravity
- Mass moment of inertia — resistance to angular acceleration

**Garbage in, garbage out:** the numbers are only right if the material density is set correctly and the model is a single watertight solid — an open or multi-body model gives a meaningless mass.



# Unit 1 Wrap-Up

1

## Controlled sketches

fully defined, with relations that capture intent

2

## Multiview drawing

correct projection, line types, and dimensions

3

## First parametric model

a clean, editable feature tree



**You'll produce:** controlled sketches, a multiview drawing, and a first fully parametric part model.

**Assessed by:** Unit 1 Quiz (Week 7) and the corresponding portfolio modules.

UNIT 2 • WEEKS 7–10

# CAD Advanced

*Detailed views, fasteners, and production drawings — turning a model into a document a shop can build from.*

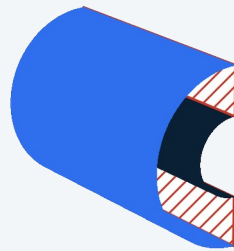
QUIZ IN WEEK 10

# Section Views

An imaginary cutting plane slices the part to show what's inside, replacing hidden-line tangles with a clear cut face.

## Full Section

The cutting plane passes entirely through the part, exposing the interior along its whole length.



## Half Section

Only one quarter is removed — the exterior and the interior appear together in a single view. Best for symmetric parts.



## Offset Section

The plane is stepped (jogged) so it passes through features that aren't in line — each still appears true size.

Top view —  
stepped cutting plane



Section A-A —  
both holes shown true size



## Local Sections

Broken-out, revolved, and removed — used to show one local feature without a full cut.

Broken-out



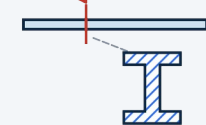
torn boundary reveals  
only the local interior

Revolved



true cross-section rotated  
90° into the view, in place

Removed



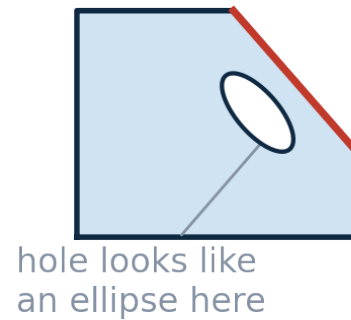
SECTION A-A

**Convention:** section lining marks the cut face — but ribs, webs, fasteners, shafts, and pins are never hatched, even when the cutting plane passes through them.

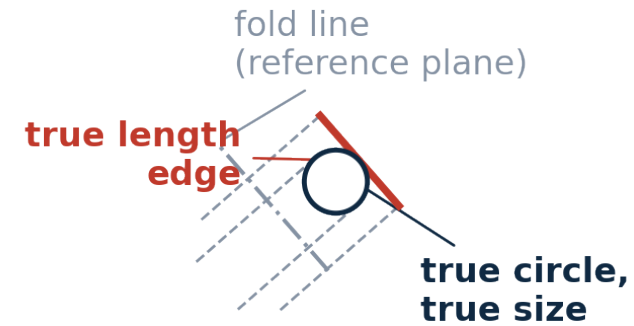
# Auxiliary Views

*An inclined face looks foreshortened in every principal view — an auxiliary view is projected perpendicular to it so it appears true-size.*

## Front view — inclined face foreshortened



## Primary auxiliary — projected true size

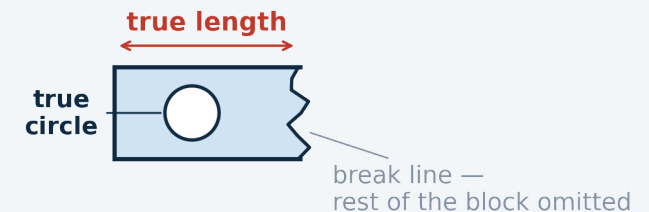


**Primary auxiliary** — projected from one principal view to show an inclined surface true-shape

**Partial auxiliary** — only the inclined feature is drawn — the rest is omitted to avoid distortion and clutter

**Reference-plane method** — measurements are transferred across the fold line along projection lines

## Partial auxiliary — only the true-shape feature is drawn

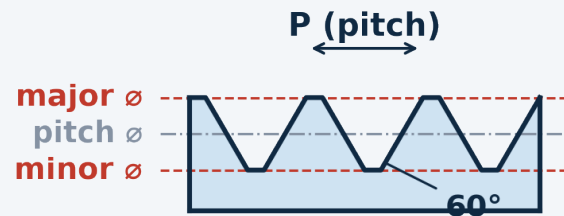


# Threads & Fastener Representation

Threads are almost never drawn in full detail — a standardized representation plus a note carries the information a shop needs.

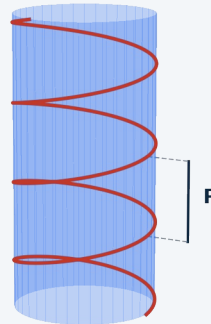
## Thread Profile

### Thread profile (external V-thread)



## How a Thread Is Cut

One helical path, advancing one pitch per turn



**Thread terms** — major/minor diameter, pitch, lead, external vs. internal, right- vs. left-hand, class of fit

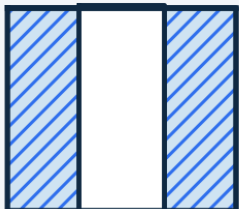
**Representation** — detailed, schematic, simplified — simplified is most common on production drawings

**Fasteners** — bolts, screws, nuts, washers, inserts — clearance and tap-drill holes are sized to fit them

**Thread notes** — 1/4-20 UNC-2A (inch) or M6 × 1 (metric) — diameter, pitch, and class in one callout

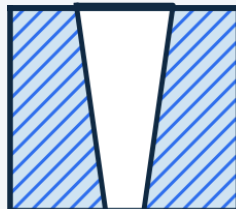
## Common Hole Types

### Through Hole



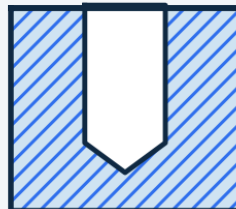
passes  
all the way through

### Tapered Hole



diameter changes  
along its length

### Blind Hole



stops before the  
far side (drill point)

### Counterbore



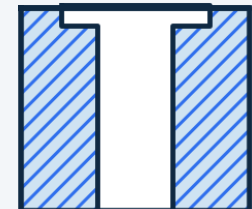
flat-bottomed step  
for a bolt head

### Countersink



conical seat for a  
flat-head screw

### Spotface



shallow flat land  
for a bearing surface

## Production & Detail Drawings

*A detail drawing is a controlled document — its format carries information that isn't in the geometry.*

[illegible]

*A blank sheet layout — every real drawing fills these same zones with the part's views, dimensions, and notes.*

# Introduction to Tolerancing

No part is made to an exact dimension — tolerancing states how much variation is acceptable so parts still fit and function.



Diameter

ø25

R

Radius

R5

±

Plus/Minus

±0.01

SQ

Square

SQ20

SR

Spherical Radius

SR10

°

Degrees

45°

25

Basic Dimension

[25]

( )

Reference Dim.

(25)

## Limit Dimensions

1.253  
—  
1.247

high limit over low limit — no math on the shop floor

## Plus/Minus Tolerance

←————→  
1.250 ±0.003

bilateral shown; unilateral: +0.005 / -0.000

## GD&T Callout



position, ø0.10 at MMC, to datum A

## Close Fit



ø25 H7/p6

near-zero gap — press or light-tap assembly (interference)

## Loose Fit



ø25 H11/c11

visible gap — free-running clearance, easy to assemble

## Surface Finish



roughness average (µin/µm) — smaller number = smoother surface

# CAD Software Configuration

*Consistent output comes from configured software, not from fixing every file by hand.*

1

## Templates

part, assembly, drawing templates with the right units, standard, title block

2

## Document properties

units, dimensioning standard (ASME/ISO), grid, layers

3

## Custom properties

file metadata that auto-populates the title block and BOM

4

## File management

naming, folder structure, keeping references intact



# CAD/CAM Data Translation

*Moving a model between systems means exporting to a neutral format — each keeps some information and loses the rest.*

## Solid exchange

*STEP (AP203/214/242), IGES*

- Geometry preserved
- Parametric feature history is not

## Kernel formats

*Parasolid, ACIS*

- Used between packages sharing a modeling kernel

## Mesh formats

*STL, 3MF*

- Faceted surface, no true curves
- Resolution set at export

**Why it matters:** lost or corrupted faces, gaps requiring geometry healing, and unit-scale errors are the common translation issues.

# Unit 2 Wrap-Up

*Freedom in modeling, discipline in documentation — how you build the part stays open-ended; the drawing follows shared conventions (ASME Y14.5, ISO) so anyone can read it.*

1

## Documentation-ready drawing

section & auxiliary views, complete title block

2

## Thread & fastener callouts

with dimensional tolerances applied

3

## Configured templates

and a clean, translated model export



**You'll produce:** a fully detailed production drawing with section & auxiliary views, fastener callouts, tolerances, plus a correctly configured template and clean export.

**Assessed by:** Unit 2 Quiz (Week 10) and the corresponding portfolio modules.

UNIT 3 • WEEKS 11-13

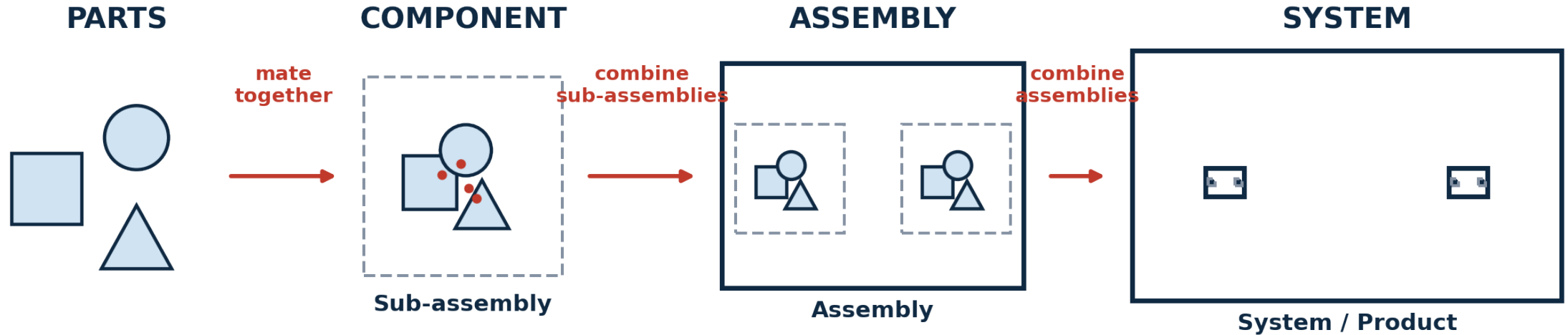
# Design for Assembly

*Combine part models into working assemblies, document them, and design them to be quick and cheap to put together.*

QUIZ IN WEEK 13

# From Parts to Systems

*The same relationship repeats at every scale: individual parts mate into a component, components combine into an assembly, and assemblies combine into a complete system.*



**Why break it down:** this is how large, complicated systems stay manageable — a vehicle or an instrument is never modeled as one flat list of thousands of parts. It's a tree of assemblies, each one a self-contained, testable unit.

# Assembly Modeling: Mates & Constraints

An assembly positions components relative to one another using mates — each removes degrees of freedom.

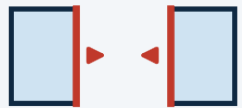
## 6

degrees of freedom a free rigid body has  
— mates remove them one axis at a time.

**First component** — the first part placed is fixed and becomes the ground the rest is built against

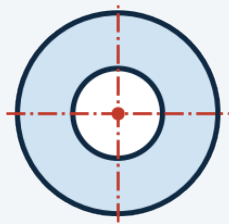
**Motion** — leaving rotational or linear freedom on purpose so a mechanism can be moved and checked

### Coincident



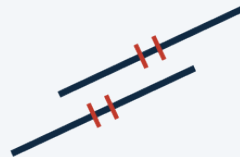
faces (or edges)  
forced flush

### Concentric



shared axis —  
pin in a hole

### Parallel



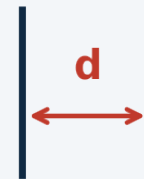
edges stay  
equidistant

### Perpendicular



edges held  
at 90°

### Distance



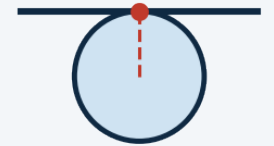
fixed gap  
between faces

### Angle



fixed angle  
between edges

### Tangent



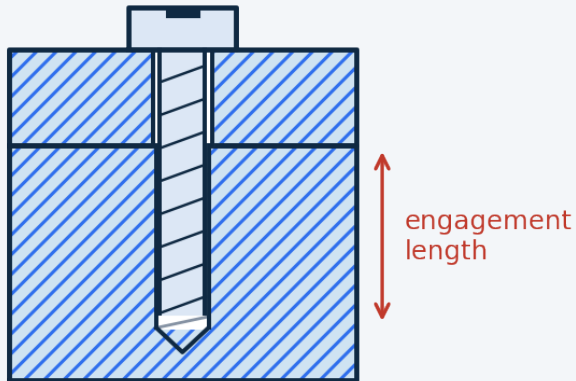
curve just  
touches a line

# Methods of Joining, Part 1

How two parts are held together is a design decision — choose deliberately, then design both parts to suit it.

## Threaded Fasteners

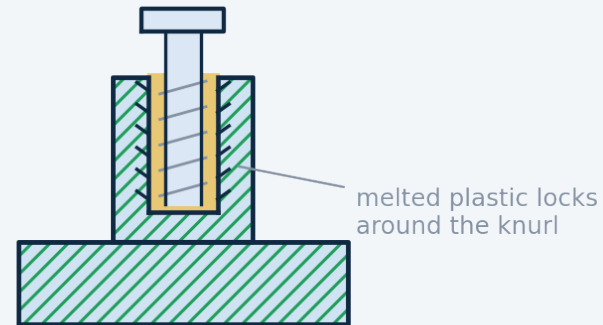
*Into tapped holes*



Default for metal-to-metal joints — strong, standard, and removable. Needs  $\sim 1.5\text{--}2\times$  screw diameter of thread engagement plus tool access.

## Heat-Set Inserts

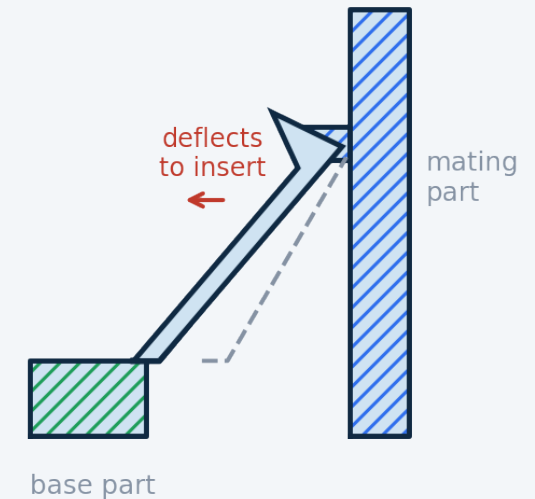
*For plastic / 3D-printed parts*



A brass insert is pressed into a boss with a soldering iron — melted plastic locks into its knurls, far more reusable than threading into plastic directly.

## Snap-Fit Joints

*No hardware, no tools*

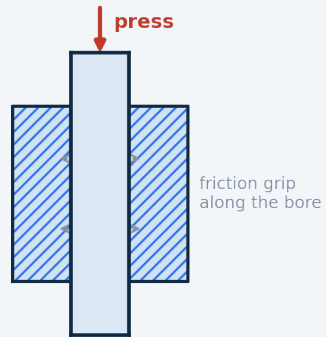


Cantilever, annular, and torsion snaps release or lock permanently depending on retention angle — design for allowable strain and a lead-in angle.

# Methods of Joining, Part 2

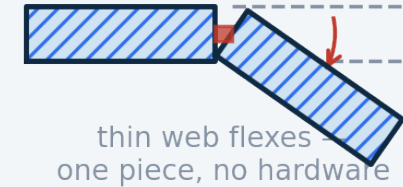
## Press / Interference Fits

Hold by friction from a slight size mismatch between mating diameters — dowel pins, bushings, bearings. Cheap and precise, but hard to disassemble and tolerance-sensitive.



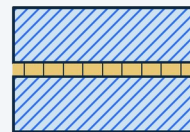
## Compliant Mechanisms

Motion or retention comes from the material's own flexibility — living hinges, flexure pivots, one-piece clips. Cuts part count to the extreme; limited travel and fatigue life.

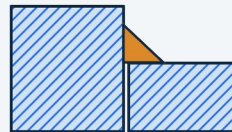


## Permanent Joints

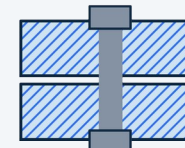
*Adhesives, solvent/ultrasonic welding, and riveting — used only where the joint never needs to separate.*



Adhesive



Weld



Rivet

**The DFA trade-off:** hardware is universal and serviceable but adds parts, tools, and assembly time; snap-fits and compliant features add almost nothing to the parts list but must be designed carefully and are harder to revise once tooling is dialed in.

# Using Online Component Repositories

*You should not model a part you can download — this is normal professional practice, not a shortcut.*

**Where models come from** — McMaster-Carr (download STEP/native from almost any product page), manufacturer sites, TraceParts, GrabCAD, 3D ContentCentral

**Why it matters for DFA** — a downloaded model is the real catalog part — correct dimensions and thread, a real part number for the BOM, at the price and lead time listed

**Bring them in cleanly** — keep in a dedicated library folder, don't edit their geometry, mate to them like any other component

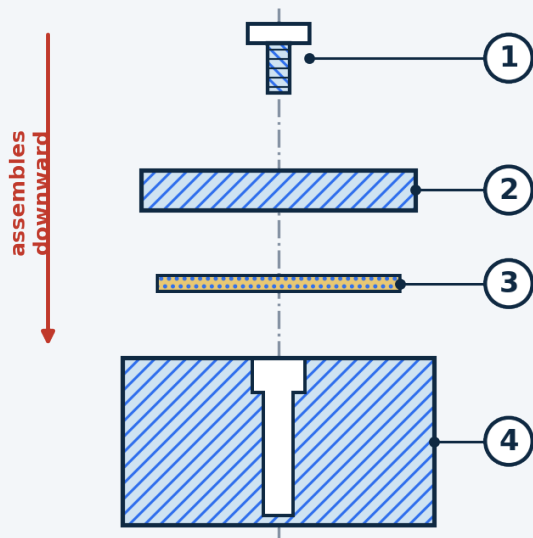
**Check before you trust** — confirm units/scale on import, simplify heavy models, verify the part number matches what you intend to buy



# Assembly Drawings

An assembly drawing shows how parts go together — exploded, numbered, and tied to a parts list.

## Exploded view — balloons keyed to the BOM



Balloon numbers are assigned in assembly order and match the item numbers in the parts list at right.

## Bill of Materials

| ITEM | PART NO. | DESCRIPTION                    | QTY |
|------|----------|--------------------------------|-----|
| 1    | HW-0425  | Socket head cap screw, M4 × 12 | 4   |
| 2    | 100-002  | Cover plate                    | 1   |
| 3    | 100-003  | Gasket, silicone               | 1   |
| 4    | 100-001  | Housing (base)                 | 1   |

In CAD, this table is generated directly from the assembly and stays linked — edit the model and the parts list updates automatically.

# Design for Assembly (DFA) Principles

*Fewer parts, easier insertion, less to get wrong on the line — DFA turns assembly time into a design variable.*

1

## Minimize part count

a part is a candidate for elimination unless it moves, needs a different material, or must be removable

2

## Prefer off-the-shelf parts

cheaper, faster, and more reliable than custom-fabricated — see the online repositories slide

3

## Easy insertion & no reorientation

chamfers and self-locating features, assembling from one direction

4

## Standardize fasteners & use symmetry

fewer unique sizes to stock and pick; make orientation obvious or irrelevant

**Design review & iteration:** the unit ends with a design review workshop — present an assembly, take feedback against the DFA principles, and produce a documented revision with a measurable improvement.

# Unit 3 Wrap-Up

1

## Multi-part assembly

correct mates, sub-assemblies, and library hardware

2

## Exploded drawing + BOM

balloons keyed to a linked bill of materials

3

## Documented revision

a design review turned into a measurable DFA improvement



**You'll produce:** a multi-part assembly model with correct mates, an exploded drawing with balloons and a linked BOM, plus a documented design revision.

**Assessed by:** Unit 3 Quiz (Week 13) and the corresponding portfolio modules.

UNIT 4 • WEEKS 14-16

# Design for Manufacturing

*Close the loop from CAD model to physical part — manufacturing-ready files, process limits, and shop safety.*

QUIZ IN WEEK 16

# Why Standards Carry the Design Intent

*Once a design leaves your screen, the maker builds exactly what the documentation says.*

**The maker is not the designer** — standard conventions (ASME Y14.5, GD&T, thread/finish callouts) let them read your intent without a phone call

**Critical dimensions must be stated** — a tolerance callout tells the shop where to spend effort and where a loose result is fine

**Standard formats preserve meaning** — STEP carries clean geometry between systems; a sloppy export arrives with gaps, wrong units, or lost features

**Ambiguity is expensive and slow** — on a graded prototype, it comes back as a part you can't use and no time to redo

**Why it matters:** capture the design intent, then communicate it in the form everyone downstream already knows how to read.

# GD&T: The Building Blocks

*A symbolic language (ASME Y14.5 / ISO GPS) for functional relationships — design intent written in symbols instead of prose.*

**Feature of size** — a feature with an axis or centerplane — a hole, pin, slot, width

**Datums (A, B, C)** — reference surfaces chosen to match how the part actually seats in use

**Datum reference frame** — primary + secondary + tertiary datums lock all six degrees of freedom

**Tolerance zone** — the geometric space the feature must lie within — a gap, a cylinder, a band

**Symbol families** — the standard's ~14 geometric characteristic symbols sort into five families — form, orientation, location, profile, and runout (next slide)

*Feature control frame — read left to right*

|                                                                                     |                             |       |
|-------------------------------------------------------------------------------------|-----------------------------|-------|
|  | $\varnothing 0.2 \text{ M}$ | A B C |
|-------------------------------------------------------------------------------------|-----------------------------|-------|

Symbol (position)

Tolerance value

Datums referenced

**Why it beats plain  $\pm$ :** a  $\pm$  box on four hole dimensions creates a small square zone and over-constrains the part; a position callout defines a round zone tied to the surfaces that matter, letting more good parts through.

# GD&T Symbol Categories

The standard's ~14 geometric characteristic symbols sort into five families by what they control and whether they need a datum.

## Form

No datum needed

 Straightness

 Flatness

 Circularity

 Cylindricity

## Orientation

Needs a datum

 Perpendicularity

 Parallelism

 Angularity

## Location

Needs datums

 Position

 Concentricity

 Symmetry

## Profile & Runout

Profile: no/some datums · Runout: datum axis

 Profile of a line

 Profile of a surface

 Circular runout

 Total runout

**Material condition modifiers:** MMC (maximum material condition) and LMC (least material condition) let a position tolerance earn bonus tolerance as the hole moves away from its worst-case size — matching how a bolted clearance-hole joint really behaves.

# Manufacturing-Ready Files

*Each process consumes a different file — a good part starts with exporting the right one at the right settings.*

## Additive

- Watertight mesh: STL or 3MF
- Fine enough to hide facets, not so fine the file is unwieldy

## Laser Cutting

- 2D vector profile (DXF)
- Cut and engrave lines on separate layers

## CNC Machining

- Solid model (STEP) + dimensioned drawing
- Drawing tells the machinist which dimensions are critical

**CAM basics** — stock setup, toolpath generation, simulation, and the post-processor that turns toolpaths into machine G-code.



# Machine Safety, PPE & Operation

*Students are checked out on safe setup and operation before running any shop equipment.*

**Baseline PPE** — ANSI Z87.1 safety glasses, closed-toe shoes, no loose clothing/jewelry/unrestrained hair

**Rotating machinery** — never wear gloves near a spindle, chuck, or bit — a caught glove pulls the hand in

**Laser cutter** — never run unattended; keep a view of the cut; know the extraction and fire response

**3D printers** — hot nozzle and bed — let them cool, don't reach into an active machine

**Before starting** — know the location of the e-stop, fire extinguisher, and first-aid kit

**PPE is not optional:** per the syllabus, a participant without proper PPE during machine use will be asked to leave the class for that session.

# Additive Manufacturing (3D Printing)

*The course uses material-extrusion printing (FDM/FFF), building one deposited layer at a time.*

**Materials** — PLA, PETG, ABS — printability, strength, and temperature trade-offs

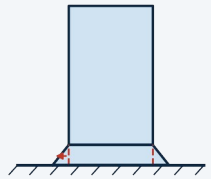
**Orientation** — layers are weakest in the build (Z) direction — a strength decision, not just fit-on-bed

**Supports & overhangs** — the roughly 45° rule, bridging limits, designing to avoid support

**Settings that matter** — layer height, wall count, infill density and pattern

**Tolerances** — holes print undersize, first layers spread — allow for it or drill/ream after

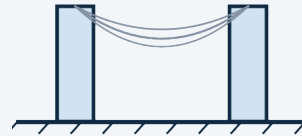
## Common Print Issues & Fixes



### Elephant Footing

*the first layer over-extrudes and squishes outward under the nozzle*

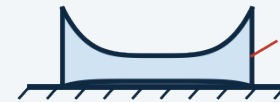
**Fix** — add a brim, or reduce first-layer horizontal expansion



### Stringing

*molten filament oozes from the nozzle while traveling between features*

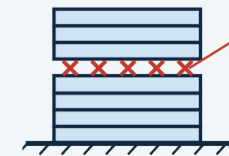
**Fix** — dry the filament; tighten retraction distance/speed



### Warping

*outer edges cool and shrink faster than the center, curling corners up*

**Fix** — add a brim/raft; enclose the printer or use a heated bed



### Layer Separation

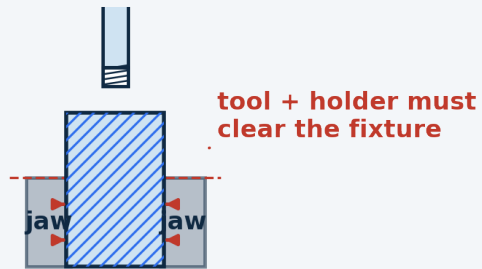
*layers don't fully fuse — often too cold, or cooled by a draft*

**Fix** — raise nozzle temp slightly; shield the print from drafts

# Subtractive Manufacturing (CNC)

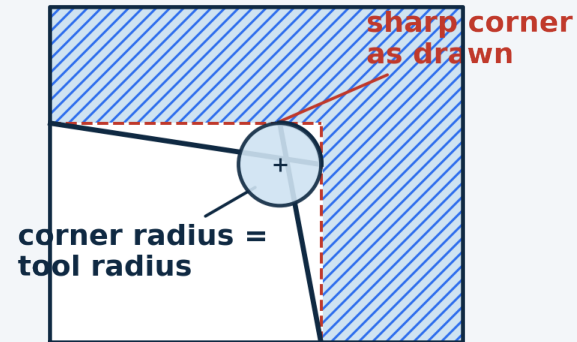
*Starts with solid stock and cuts material away — design constraints come from the tool and how the part is held.*

## Workholding: rigid, and clear of the tool



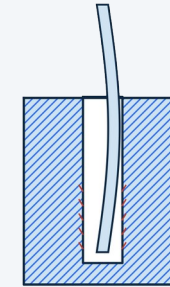
workpiece clamped between vise jaws

## Internal corners inherit the tool radius



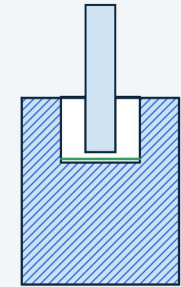
## Tool reach vs. stiffness

Deep + narrow:  
long tool deflects



long, thin tool flexes  
under load — chatter &  
tapered walls

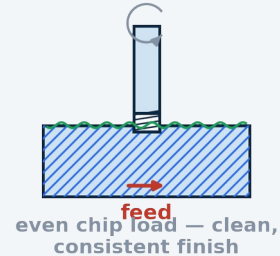
Shallow + wide:  
short tool stays stiff



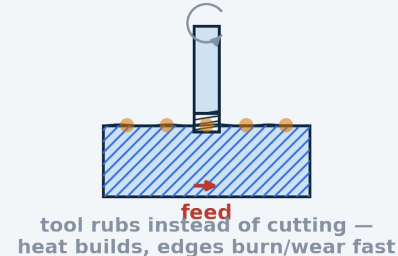
short, stiff tool —  
straight walls,  
clean finish

## Feeds & Speeds

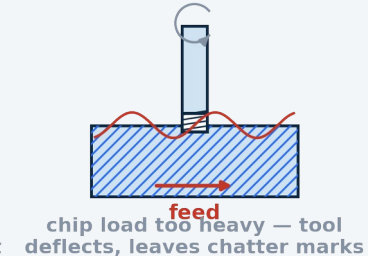
### Balanced



### Feed too slow



### Feed too fast



# Laser Cutting

*Cuts flat sheet by burning or vaporizing a narrow line through it — fast and precise in 2D, with its own rules.*

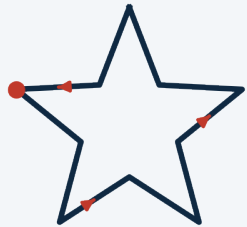
**Kerf** — the cut removes a finite width of material — account for it on tight-fitting parts

**Material limits** — wood, acrylic, paper, and cloth cut well

**Small-feature limits** — minimum hole size and spacing relative to material thickness

## Vector Cut vs. Raster Engrave

Vector cut



single continuous path —  
cuts through along the outline

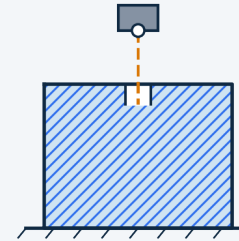
Raster engrave



many parallel scan lines —  
burns the surface, fills the area

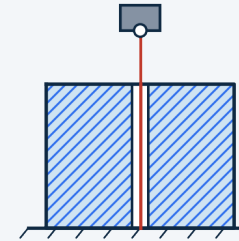
## How Laser Power Shows Up in the Cut

Power too low



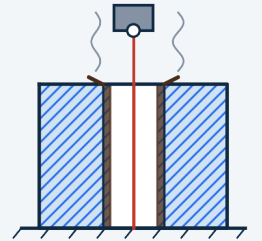
doesn't cut through —  
needs multiple passes

Power well-tuned



clean, narrow kerf —  
through in one pass

Power too high



wide, charred kerf —  
melted edges

**Never cut:** PVC and other chlorinated plastics must never be cut on a laser — they release chlorine gas.

# DFM Design Considerations

*Shape the part so the chosen process can make it well, at reasonable cost, without special effort.*

1

## Match design to process

uniform walls for printing/casting, generous radii for machining, flat unfolded geometry for laser

2

## Minimize setups

features on one face are cheaper than features needing re-fixturing

3

## Tolerance only what matters

tight tolerances drive cost — apply to mating/functional features

4

## Use standard stock & tooling

standard sizes and drill/tool radii cost less, arrive faster

5

## Design for prototype AND product

what's fine for a one-off print may not survive production

# Unit 4 Wrap-Up

1

## Manufacturing-ready files

correctly exported for the chosen process

2

## GD&T documentation

for at least one process, showing what you'd control and why

3

## A DFM-reviewed design

ready to hand off to a shop process



**You'll produce:** manufacturing-ready files and GD&T documentation demonstrating your design-for-manufacturing reasoning.

**Assessed by:** Unit 4 Quiz (Week 16). The physical prototype build & presentation is graded separately as the course Final Exam.

# All Units at a Glance

Keep this page open while you work — every unit's timing, focus, and assessment in one place.

|                                     |               |                                                                    |                    |
|-------------------------------------|---------------|--------------------------------------------------------------------|--------------------|
| ● Foundations                       | Before Unit 1 | Define the problem, design intent, top-down vs. bottom-up          | —                  |
| ● Unit 1 — CAD Fundamentals         | Weeks 4–7     | Controlled sketches, multiview drawing, first parametric model     | Quiz, Wk 7         |
| ● Unit 2 — CAD Advanced             | Weeks 7–10    | Section/auxiliary views, threads, tolerancing, production drawing  | Quiz, Wk 10        |
| ● Unit 3 — Design for Assembly      | Weeks 11–13   | Mates, joining methods, exploded drawing + BOM, DFA principles     | Quiz, Wk 13        |
| ● Unit 4 — Design for Manufacturing | Weeks 14–16   | GD&T, manufacturing files, additive/subtractive/laser, DFM, safety | Quiz, Wk 16        |
| ● Final Exam — Prototype Project    | Week 17       | Team-built physical prototype, manufacturing docs, presentation    | 30% (w/ prototype) |